

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-acb5895c0dd22d767.jsonl`  
- SHA-256: `c3eacf4eab0102fe59247b845e99115040b08cc4671ac5906cc031df90b6c80b`  
- Source modified: `2026-06-22T01:58:05+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_6abc9156-811`  
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T01:55:49.365Z)

You are scanning planning docs to surface ACCURATE, prioritized next steps for a co-founder.

REPO RULES:

```
- Engine repo at /c/Users/avidu/Projects/badminton-highlight-indexer: its working tree is on a STALE serving branch. Read DOCS + verify SHIPPED state from ENGINE origin/master ONLY:  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer show origin/master:docs/FILE.md  
(read a doc on master; if it errors, the doc may be branch-local ? fall back to the working-tree file and SAY SO)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60  
(what actually shipped ? use this to catch stale "TODO/next" claims)  
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "PATTERN" origin/master  
-- 'PATHGLOB'  
- khelsutra repo at /c/Users/avidu/Projects/khelsutra: on clean master ? use Read/Grep/Glob directly + git -C /c/Users/avidu/Projects/khelsutra log --oneline -40.
```

ANTI-SPIRAL (critical): before listing any "[gap]/net-new/TODO/next" item, VERIFY it hasn't already shipped ? cross-check the doc's claim against git log origin/master. A doc saying "X is next" is a HYPOTHESIS until you confirm X isn't already merged. Flag any stale doc claims you correct.

Context (already TRUE as of this session, do not re-list as "next"): the khelsutra ALPHA is DEPLOYED + live on a CPU droplet (engine + cloudflared tunnel + Cloudflare Access + SPA on Pages, real Gemini key working, full chain verified). DEPLOY_ALPHA fixes + the droplet kit are merged (khelsutra #78/#79/#81/#85). T2 (alpha deploy) is effectively DONE.

Return prioritized next items with honest status + git grounding.

TRACK: ****packaging / downloadable app**** (the co-founder explicitly asked: "can I download a package and run it on a new machine now?").
Read ENGINE `docs/PACKAGING\_PLAN.md` (origin/master) end-to-end + verify the P-series PRs against `git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60` (look for P0/P1/P2 packaging commits, `indexer --app`, "bundle the SPA", "lazy-torch", "wheel").
ANSWER PRECISELY: (1) what works TODAY for "download + run on a new machine" ? is it `pip install` + `indexer --app`? a built wheel? a double-click installer? Verify by checking the console-script entrypoint + the bundled-SPA serving (grep origin/master for the entrypoint + `--app`). (2) the EXACT remaining gap to a non-technical user downloading & running it on a fresh machine (the P2b-install / P2c-wizard items ? confirm they're still open vs merged). (3) prioritized packaging next items.

2. assistant (2026-06-22T01:55:52.491Z)

I'll scan the packaging docs and verify against the engine's origin/master. Let me start by

pulling the latest and reading the key sources.

3. user (2026-06-22T01:55:57.174Z)

---LOG---

a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export) (default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale (#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place) (default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-OFF) (#302)
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities, /api/videos, /api/reels (#300)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes (#297)
3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard (default-OFF) (#296)
fc78fce feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config (#295)
0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF) (#294)
82a28f4 feat(pkg): P0a ? resolved app-data HOME (RALLY_APP_HOME) for config/.env/DB/output (default-OFF) (#293)
2afbadf fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)
7835f34 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide (#291)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
fefa412 docs(serving): resolve S8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project (#275)
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/

```
(#276)
03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh)
(#274)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)
bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot)
(#271)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($0) (#267)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
4d546ec fix(gcp): teardown defaults to DELETE (a stopped VM still bills its disk) (#265)
dd30e9b feat(workspace): per-video workspace convention ? design doc + Phase 0 helper (#264)
dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving
model (#263)
d8c5aa0 docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state
(#262)
0d73d96 docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent
runbook (#261)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train,
GCP=serve (ADR-011) (#260)
36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a
tolerance (#259)
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
55ae408 docs(decoupled): GCP pivot runbook + DO?GCP ADR-010 (deploy/gcp + handoff) (#257)
dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction
(#256)
5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)
d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock
(#254)
814d813 feat(deploy): GPU machine provisioning ? NVMe scratch + native WASB setup (#253)
f06c6a6 fix: cross-platform canonicalization (Linux/macOS, not just Windows) (#252)
6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)
d292854 docs: SETUP_NEW_MACHINE ? canonical zero?train/infer runbook (#250)
6878d45 feat(storage): pluggable StorageBackend ? local + S3/GCS first-class buckets (#249)
```

4. user (2026-06-22T01:55:57.485Z)

Batteries-Included App ? Packaging & Distribution Plan

```
> **Status:** `IN PROGRESS` ? 01/02 resolved (LIGHT-first; cross-platform webserver, no native
component; **WASB-complete north star**); **11 PRs merged 2026-06-21/22** (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e); 03+ owner/counsel-gated . **Owner:** Avi Dullu . **Created:**
2026-06-21 . **Last updated:** 2026-06-22
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** $7 progress tracker is the source of truth; indexed in `docs/README.md`
+ `docs/NEXT_STEPS.md`; memory `[[packaging-app-plan]]` + `[[next-session-packaging-app]]`.
===EXIT 0===
```

5. assistant (2026-06-22T01:56:00.545Z)

The PACKAGING_PLAN.md exists on origin/master. Let me read it end-to-end and verify the key endpoint claims.

6. user (2026-06-22T01:56:01.580Z)

<persisted-output>

Output too large (30.3KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\tool-results\bedmy6zjd.txt

Preview (first 2KB):

Batteries-Included App ? Packaging & Distribution Plan

```
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component; **WASB-complete north star**); **11 PRs merged 2026-06-21/22** (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e); 03+ owner/counsel-gated . **Owner:** Avi Dullu . **Created:**
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+ `docs/NEXT_STEPS.md`; memory `[[packaging-app-plan]]` + `[[next-session-packaging-app]]`.
> **Relation to existing docs:** **supersedes** the narrative of `docs/DESKTOP_APP_PLAN.md` (per
locked D10 ? self-hosted webserver, not a desktop app); **builds on**
`docs/COMPUTE_DECOUPLED_SERVING/` (the profile/storage/compute seam) and is the **distribution
layer** under `kheIsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md`.
> **Honesty note:** claims tagged `[verified]` (checked at file:line), `[researched]` (web/needs
sign-off), `[design]` (proposed). Not legal advice ? licensing items need counsel.
```

? NEXT SESSION ? start here

```
**You are here (2026-06-22):** `origin/master` has the **LIGHT-tier engine fully bundle-ready**
? `pip install` ? `indexer --app` boots and serves the real KheIsutra SPA single-origin,
**torch-free**, with resolved app-home state, an env-overridable ffmpeg, DNS-rebinding-guarded
localhost binding, single-instance locking, and safe DB upgrades. 11 PRs merged; 0 open PRs; 0
GCP VMs ($0). Ramp up via this doc's $7 tracker + memory `[[packaging-app-plan]]`.
```

```
**Remaining work, in priority order (each its own small PR; branch off `origin/master`, explicit
paths ? shared checkout):**
```

...

</persisted-output>

7. user (2026-06-22T01:56:05.345Z)

```
1      # Batteries-Included App ? Packaging & Distribution Plan
2
3      > **Status:** `IN PROGRESS` ? 01/02 resolved (LIGHT-first; cross-platform webserver, no
native component; **WASB-complete north star**); **11 PRs merged 2026-06-21/22** (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e); 03+ owner/counsel-gated . **Owner:** Avi Dullu . **Created:**
2026-06-21 . **Last updated:** 2026-06-22
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md` + `docs/NEXT_STEPS.md`; memory `[[packaging-app-plan]]` + `[[next-session-
packaging-app]]`.
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on** `docs/COMPUTE_DECOUPLED_SERVING/` (the profile/storage/compute seam) and is the
**distribution layer** under `kheIsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md`.
7      > **Honesty note:** claims tagged `[verified]` (checked at file:line), `[researched]`
(web/needs sign-off), `[design]` (proposed). Not legal advice ? licensing items need counsel.
8
9      ---
10
11     ## ? NEXT SESSION ? start here
12     **You are here (2026-06-22):** `origin/master` has the **LIGHT-tier engine fully bundle-
ready** ? `pip install` ? `indexer --app` boots and serves the real KheIsutra SPA single-origin,
**torch-free**, with resolved app-home state, an env-overridable ffmpeg, DNS-rebinding-guarded
localhost binding, single-instance locking, and safe DB upgrades. 11 PRs merged; 0 open PRs; 0
GCP VMs ($0). Ramp up via this doc's $7 tracker + memory `[[packaging-app-plan]]`.
13
14     **Remaining work, in priority order (each its own small PR; branch off `origin/master`,
explicit paths ? shared checkout):**
15     1. **P3 GPU validation + screencast (owner wants first; ~$0.50, well under the $100
cap).** Provision an L4 (`deploy/gcp/create_gpu_vm.sh`) ? install the **wheel** on the fresh GPU
box + the WASB env ? validate the packaged stack end-to-end (`indexer --app` ? bundled SPA ?
```

native-WASB ? reel) ? ****screencast**** ? `teardown.sh` (deletes) + confirm \$0. ? Run as a FOCUSED op (billable VM ? teardown is priority #1). ? Known blocker: native-WASB returned ****0 rallies**** on a fresh build last session ? reproducing the 22-rally result needs the engine session's exact tuned `worker infer detector\_impl=native` config. Confirm cmd + cost before spend; `[[gcp-vm-always-delete]]`.

16 2. ****P2b-install**** (\$7) ? `uv tool install` / a generated cross-platform install script (now a clean ****torch-free**** LIGHT install thanks to P2e) + ship a truly-local default `config.json` + an uninstall manifest.

17 3. ****P2c first-run wizard**** (\$7) ? the 3-step wizard UI (Gemini key / compute / videos) in the ****khelsutra `web/` SPA****, rendered from `GET /api/capabilities`. ****Cross-repo**** (`avidullu/khelsutra`) ? coordinate like P1b (the SPA is gitignored-dist; refresh `backend/ui/` via `scripts/bundle\_ui.sh` after).

18 4. Then ****P3-cloud**** (Cloud Run dispatch, micromamba Dockerfile ? P3a) and ****P4**** (ONNX export ? dissolve the dual-env; training + in-app annotation = the ****north-star**** completion).

19

20 ---

21

22 ## 0. TL;DR

23 Ship the engine as ****ONE pip-installable artifact** whose behaviour is selected entirely by the already-merged decoupled-compute seam (`deployment.profile` / `compute\_target` / Storage+Compute backends), fronted by a friendly launcher + a Windows installer ? ****not**** a frozen monolith, ****not**** an Electron/Tauri shell. The hard constraint that forces scope: torch is deliberately out of `pyproject` (needs `--index-url`) ****and**** native WASB needs a **second, incompatible** torch env (conda py3.8 + torch 1.11+cu113) shelled out via `\$WASB\_PYTHON` `[verified native\_wasb\_runner.py:89]` ? so no single venv/freeze can deliver the GPU-quality path. ****The honest v1 is a LIGHT tier**** (Gemini + motion + cpu-mock, **zero** torch/GPU) that preserves CUJ-1.5/8; the native-WASB GPU path (CUJ-6/7) ships ****owner-only**** until two blockers clear (WASB-C3 weight provenance + a reproducible truly-local config). ****LIGHT is the FIRST shippable increment, not the terminal scope ? the sole completion goal (north star, L10) is the FULL quality path: native WASB + BYO training + in-app annotation.**** Delivery is ****cross-platform** (Windows/macOS/Linux): the same local webserver powers the UI on all three, with NO native component (L9). ****Five small engine prerequisites must land before any packaging push**** (\$7 P0). The single most actionable finding: ****all mutable state is CWD-relative today and `--setup`/server disagree on `config.json` ? a silent data-loss footgun that P0 must fix first.**

24

25 ## 1. Problem & goal

26 ****Goal:**** a non-technical user (the primary persona: an India coach/parent on Windows ? `docs/UI\_PROPOSAL.md` §2) ****downloads and runs**** the indexer ****without sourcing the repo****. The app spins up a ****local web service**** for (a) inference video?rallies?reel, (b) BYO ****training****, and (c) ****rally annotation**** if needed ? runnable ****locally and against cloud services**** ? and ****preserves every completed CUJ**** (§ CUJ map).

27

28 ****Why now:**** scaffolding + the decoupled-compute serving seam (M1?M8) are merged and default-OFF; the SPA + i18n + the dry-run CUJ are done. What's missing is the **distribution** story: today the only path is `git clone` ? `pip install -e .` ? out-of-band torch ? manual ffmpeg ? set up the WASB conda env ? build the SPA from a **different repo**. This plan collapses that into download-and-run.

29

30 ****Read to get current:**** `pyproject.toml`, `backend/main.py` (`main()` + `--server`), `backend/config/{loader,presets,models}.py`, `backend/storage/\*`, `backend/compute/\*`, `backend/pipeline/detectors/native\_wasb\_runner.py`, `deploy/cloudrun/Dockerfile`, `docs/COMPUTE\_DECOUPLED\_SERVING/README.md`, `khelsutra/web/` + `khelsutra/docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md`.

31

32 ## 2. Decisions locked (do not relitigate)

33

34 | # | Decision | Source / date | Implication for packaging |

35 |---|-----|-----|-----|

36 | L1 | ****Pure DETECT?WINDOW?STITCH core never branches on profile****; one build, profile-selected | COMPUTE_DECOUPLED_SERVING D1 | Local vs cloud is ****one config flip****, not a fork ? build ON this, don't reinvent. |

37 | L2 | ****Profiles**** = truly-local / cloud-serving / cpu-mock via `PROFILE\_PRESETS` | M1 #279 `[verified backend/config/presets.py]` | The app exposes a single "where does compute run" choice. |

```

38 | L3 | **Compute = GCP only** (Cloud Run GPU Jobs); DigitalOcean dropped | ADR-013,
2026-06-20 | Cloud tier targets Cloud Run + BYO-GPU-VM only. |
39 | L4 | **Edge-auth required before any non-localhost exposure** (Cloudflare Access + in-
app JWT verify) | D9 / M9 `[verified backend/api/auth.py wired default-OFF main.py:294]` | Bind
stays `127.0.0.1` by default; `0.0.0.0` only under an edge-auth profile. |
40 | L5 | **Delivery = self-hosted webserver the user spins up ? NOT a desktop app** | D10,
2026-06-20 (supersedes `DESKTOP_APP_PLAN.md`); **01 confirmed 2026-06-21** | The launcher boots
the server + opens the user's *own browser* at localhost ? that IS the delivery, no embedded
webview. |
41 | L6 | **MIT / Apache-2.0 / BSD only** for every dependency ? code, data, AND weights |
ratified 2026-06-09 | Every *new packaging* dep must clear this (see §3). |
42 | L7 | **Paid LLMs (Gemini) = inference/serving only, never a training data source**;
commercial models trained on paid-LLM labels are ship-blocked | guardrail + `gate_verdict`
`[verified harness.py:177 ship=False]` | The wizard's Gemini-key step must say so; training UI
must surface the ship-gate. |
43 | L8 | **User-switchable profiles** ("your machine or our cloud") with explicit confirm
? cloud spend never silent | D13 / D15 | "Push to cloud" toggle needs confirm + INR estimate
from `backend/billing.py`. |
44 | **L9** | **Cross-platform (Windows + macOS + Linux); NO native component** ? the same
local webserver powers the UI on all three | 01, 2026-06-21 | Distribution = `uv`/pip + a
**pure-Python launcher** (`webbrowser.open`). Reject native-installer-as-runtime,
Electron/Tauri, and LGPL tray libs. OSes differ only in ffmpeg/torch *acquisition*. |
45 | **L10** | **North star = the FULL quality path is the SOLE completion goal** ? LIGHT
is the first increment; DONE requires native WASB + BYO training + in-app annotation | 02,
2026-06-21 | §9 DoD splits "v1 shippable" from "north-star DONE"; P3/P4
(WASB/ONNX/train/annotate) are completion-gating, not optional. |
46
47 ## 3. Foundation ? research / prior art
48
49 **Distribution channels `[researched]` **`uv` is the only channel that can both
install a pinned Python *and* route torch's `--index-url`; `pipx`/plain `pip` can't express the
torch index. Frozen binaries (PyInstaller/Nuitka) with torch+CUDA are 3?5 GB, can't split CUDA
libs, and **cannot host the second WASB env** ? rejected for the GPU path. Prior art (Frigate
NVR ? closest analog: ffmpeg + ML + web + Docker; Ollama; ComfyUI; LM Studio) converges on **two
patterns that fit us: **(1) a **container image** for the server/GPU profile, (2) a **first-run
download + a friendly launcher ** for the local app.
50
51 **Licensing landmines `[verified]` (the load-bearing §3 findings): **
52 - **WASB-C3 is the dominant blocker.** The WASB-SBDT *code* is MIT, but the badminton
*checkpoint* provenance is **unresolved** for commercial redistribution ? `gate_verdict`
hardcodes `ship=False` `[verified training/gen0/harness.py:174-177]`. **Do not conflate MIT-code
with cleared-weights. ** A public download must **not** auto-fetch the weights until counsel
signs off (03).
53 - **ffmpeg is GPL via the `libx264` encoder** `[verified
backend/pipeline/stitcher/compiler.py:253]`. The license-clean path is **not to redistribute
ffmpeg** ? resolve a system/winget install instead. `imageio-ffmpeg` also lacks `ffprobe` (which
the engine needs) `[researched]`.
54 - **Anaconda ToS exposure:** the Cloud Run image uses `Miniconda3` + `conda tos accept ?
pkgs/main`/`pkgs/r` `[verified deploy/cloudrun/Dockerfile:33-37]` ? the ?200-employee ToS trap.
Replace with **micromamba + conda-forge** (BSD, no ToS) for the cloud image *and* any local
managed WASB env; record an ADR.
55 - **New packaging deps that clear L6 `[researched]` **`uv` (Apache/MIT), Inno Setup
(permissive), `onnxruntime` (MIT), `micromamba` (BSD-3), Docker/OCI (Apache-2.0), boto3/google-
cloud-* (Apache-2.0), stdlib `webbrowser`. **Forbidden:** `pystray` (LGPL-3.0). **Build-time-
only: **`Nuitka` 4.x is **AGPL-3.0** (compiled output has a runtime exception) ? usable as a
tool, never bundled. `obsreport` (private git dep) stays a truly-optional extra, never bundled.
56
57 ## 4. Design / architecture
58
59 ### 4.1 One build, profile-selected (the reuse map)
60 - **Reuse:** hatchling build + `indexer = backend.main:main` console script `[verified
pyproject.toml:80-82]`; `PROFILE_PRESETS` `[verified presets.py]`; `get_compute_backend(config)`
returns `CloudRunCompute` only for `compute_target='cloud_run'` else `None`=in-process
`[verified backend/compute/__init__.py]`; `backend/storage/*` (local/gcs/s3/gdrive); the in-
process `ThreadPoolExecutor(max_workers=1)` job worker `[verified backend/api/job_worker.py]` ?

```

****server and worker are the same process for serving**** `[verified main.py:729-736]`, so a packaged launcher just runs `indexer --server`.

61 - ****New:**** the app defaults to ****truly-local****; the user toggles profile in Settings (rendered from a new `/api/capabilities`) with explicit confirm (L8) + INR estimate before any cloud job.

62

63 **### 4.2 How one build runs local + cloud**

64 The local FastAPI server is the ****control plane in every profile.**** `truly-local` runs detect/stitch in-process. `cloud-serving` flips `compute\_target=cloud\_run` ? the local control plane dispatches a ****Cloud Run GPU Job**** (the **same** image, forced inline) and bucket-polls a done-marker; `storage=gcs` routes I/O. `cpu-mock` gives a \$0 deterministic self-test on first run.

65

66 **### 4.3 How the UI is bundled (collapses the cross-repo + reverse-proxy problem)**

67 Build `khelsutra/web` (`npm ci && npm run build`) ****at package time**** and vendor `web/dist/` into the wheel via a ****hatchling force-include**** ? a package dir (e.g. `backend/ui/`), served by FastAPI with an ****SPA history-fallback**** route. The SPA already calls ****same-origin `BASE='/api'`**** `[verified khelsutra/web client.ts:7]` ? ****zero SPA code change**** for single-origin serving. Retire the legacy CWD-relative `backend/api/static` mount `[verified main.py:71-72]`; resolve paths via `importlib.resources`. `dist/` is gitignored ? the build must record a ****bundle hash**** + an `/api/version` handshake + a contract test pinning SPA?engine.

68

69 **### 4.4 Dual-torch / weights / ffmpeg ? concretely**

70 - ****Dual-torch:**** do ****not**** host both ABIs in one venv or freeze. v1 LIGHT carries ****no torch at all****. The GPU/WASB HEAVY tier is a ****separately-provisioned second env**** ? short term via ****micromamba + conda-forge**** reached by `\$WASB\_PYTHON` exactly as `native\_wasb\_runner.py:89` already does; ****strategic fix = export WASB to ONNX**** (the `DetectorRunner` ABC already names an `ONNXRunner` writing the identical `Frame,Visibility,X,Y` CSV) so the WASB path needs **neither** torch 1.11 nor 2.x ? `onnxruntime` (MIT, tens of MB, CPU+CUDA+DirectML+CoreML) dissolves the dual-env. Owner-gated on a real-GPU byte-parity re-validation.

71 - ****Weights:**** never bake `wasb\_badminton\_best.pth.tar` into the artifact (size + C3). Fetch on first run, ****SHA-256-pinned + resumable****, into the app-data cache, setting `\$WASB\_WEIGHTS`. ****Until C3 clears, the public download must NOT auto-fetch ? owner-only flag.****

72 - ****ffmpeg:**** route ****every**** `ffmpeg`/`ffprobe` call through ****one resolver**** (config path ? env ? `shutil.which` ? fail-loud); first-run setup (extend `scripts/doctor.py`, which already prints the winget command) detects-and-offers `winget install Gyan.FFmpeg`. An LGPL ffmpeg bundle (BtBn,openh264/NVENC) is a documented fallback that also requires swapping the `libx264` default ? tracked, not v1.

73

74 **### 4.5 Tiers**

75 | Tier | Audience | Contents | Install UX |

76 |-----|-----|-----|-----|

77 | ****LIGHT**** (v1 first increment) | Non-tech India coach/parent (Win/Mac/Linux); any CPU box | wheel + bundled SPA + Gemini/motion/cpu-mock; ****no torch/GPU/weights****; ffmpeg via per-OS offer; defaults truly-local; user-supplied Gemini key | ****Cross-platform, no native component:**** `uv tool install` / a generated install script + a ****pure-Python launcher**** (`indexer --app` ? boot server ? `webbrowser.open 127.0.0.1:8000`, free-port fallback). Optional thin per-OS shortcut (.bat/.command/.desktop) wrapping the same command. First run: cpu-mock self-test ? 3-step wizard. |

78 | ****HEAVY / local-GPU**** (owner + opt-in power user) | Owner w/ golden corpus + CUDA box | LIGHT + micromamba WASB env (py3.8/torch1.11) via `\$WASB\_PYTHON`; main torch out-of-band via uv; weights SHA-256-pinned. ****Gated on C3 + reproducible config****; ONNX supersedes this env | opt-in "Enable local GPU" step; resumable progress (multi-GB on Indian bandwidth); post-install `torch.cuda.is\_available()` validate; falls back to LIGHT. |

79 | ****CLOUD / server**** | GCP deploy + BYO-GPU-VM | the `deploy/cloudrun/Dockerfile` image (micromamba-ified); cloud-serving + gcs; behind Cloudflare Access | `gcloud run jobs create` per `deploy/cloudrun/README.md`, or `deploy/gcp` VM scripts. Bind?`0.0.0.0` (config-gated) + edge-auth mandatory. |

80

81 **## 5. Threat model / risk table (local appliance)**

82 | Asset | Protected? | Residual risk / control |

83 |-----|-----|-----|

84 | Gemini API key | Partially | Today `GET /api/config` returns the ****whole `AppConfig` unauthenticated**** `[verified main.py:144-146]`. ****P0**** redacts + adds a serialization test + a Host-header/localhost-origin guard (DNS-rebinding vector against `127.0.0.1`). Key lives in app-

```

home `.env` (or OS keychain), never in config; scrubbed on uninstall. |
85     | User videos + **human rally labels** | Yes (local) | MD5 video?label join breaks on
re-encode ? never re-encode after annotating; add export/backup (P2d). |
86     | Cloud spend | Yes | Profile flip needs explicit confirm + INR estimate + per-video cap
(L8); never silent. |
87     | SQLite index | Yes | DB migration on upgrade: backup-before-migrate + downgrade guard
(P2d). |
88
89     ## 6. Honest limits ? what this does NOT do
90     - The **first increment** (LIGHT) **does not run native WASB** ? CUJ-6/7 is absent from
the *initial* download (CPU/Gemini quality ? GPU-WASB quality). Per **L10 this is temporary, not
terminal ? the north-star completion goal explicitly includes native WASB + training +
annotation (P3/P4). Don't market the first increment as the finished product.
91     - A green offline suite does **not** prove the *built artifact* works (hidden imports,
namespace-package resolution with no `backend/___init__.py`, missing data files like fonts) ? P0
adds a built-artifact smoke lane.
92     - "Push to cloud" is **BYO-GCP** today (no hosted-Khelsutra Storage/Compute client
exists) ? D13 marketing currently overstates a hosted option (O4).
93     - Training is **not** a closed loop: `worker train` is an offline CLI, `gate_verdict`
refuses commercial ship, and in-process jobs don't survive restart `[verified main.py:646
fail_stale_jobs]`. Never advertise annotate?train as done.
94
95     ## 7. Deliverables & progress tracker ? source of truth
96     Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, branched
from `origin/master`, no stacking, default-OFF.
97
98     | ID | Deliverable | Depends on | Gated? | Status | PR |
99     |----|-----|-----|-----|-----|----|
100    | **P0a** | **App-data HOME** (`RALLY_APP_HOME` env + per-OS default via
`default_os_app_home`) threaded through `default_config_path()`,
`load_dotenv(env_file_to_load())`, `resolve_output_dir` (output/DB), package-relative static
mount; **unified `doctor.py` write with the server read**. Byte-identical when unset; 4-lens
adversarial review folded. | ? | No | ? | [#293](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/293) |
101    | P0b | **Redact `GET /api/config`** (`backend/utils/redact.py`) + serialization-safety
test (incl. a full-`AppConfig` over-masking guard). *(Host-header guard moved to P0c ? same
exposure concern.)* | ? | No | ? | [#295](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/295) |
102    | P0c | **Config/profile-driven bind + DNS-rebinding Host-header guard**
(`TrustedHostMiddleware` + `_resolve_bind_host`; default `127.0.0.1`;
`0.0.0.0`+`RALLY_ALLOWED_HOSTS` under edge-auth) + a startup WARN for the edge-auth/allowlist
footgun. Byte-identical default; adversarial review folded. | ? | No | ? |
[#296](https://github.com/Khelsutra/badminton-highlight-indexer/pull/296) |
103    | P0d | **Declare `scipy`** in `[project.dependencies]` (serving use `[verified
features/rally_seq.py:17]`, was transitive-only) + **license-scan CI lane** (pip-licenses; fails
on GPL/AGPL/LGPL/SSPL) + **built-wheel smoke lane** (build ? install ? `indexer --help` from a
non-repo cwd). Both lanes green on CI. | ? | No | ? |
[#297](https://github.com/Khelsutra/badminton-highlight-indexer/pull/297) |
104    | P0e | **Single-instance OS lock** (`backend/utils/single_instance.py`, fcntl/msvcrt)
on `**? P0 (engine-prerequisite tranche) COMPLETE ? P0a?P0e all merged 2026-06-21.** Each
isolated, default-OFF, adversarially reviewed, full-suite + 9-lane CI green; coverage held ~85%.
The engine is now safe to launch from an arbitrary directory and to bundle. **? NEXT = P1**
(bundle the SPA single-origin + the 3 net-new endpoints `api/capabilities`, `api/videos` list,
`api/reels` + `api/version` contract test).
107    | **P1a** | **Net-new endpoints** (engine-only): `GET /api/version` +
`api/capabilities` (segmenters/ffmpeg/nvidia-smi/gemini-present/weights/profile ? presence-
bools only) + `api/videos` (list; new `Database.list_videos`) + `api/reels` (+
`api/reels/{filename}` download; **global flat list** ? reels have no per-video assoc today,
honest scope). | ? | No | ? | [#300](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/300) |
108    | P1b | **Bundle the SPA single-origin**: source-agnostic `scripts/bundle_ui.sh` (build

```


`khelsutra/web`@pinned now; unpack a release tarball later ? option 3) ? committed `backend/ui` snapshot (option 1, ~700KB) + `ui\_version.json` + `THIRD\_PARTY\_NOTICES.md`; hatchling auto- includes it (no force-include); FastAPI serves `/` + `/assets` (legacy fallback); `/api/version` reports the bundled UI; SPA?engine skew-guard contract test; `backend/ui/\*\* -text`. | P1a | No (cross-repo) | ? | [#302](https://github.com/Khelsutra/badminton-highlight-indexer/pull/302) | 109

110 *** P1 COMPLETE (P1a #300 + P1b #302) ? the engine now serves the real KhelSutra SPA single-origin from the wheel, with the console/wizard endpoints in place. ? NEXT = P2** (cross- platform launcher `indexer --app` + first-run wizard rendered from `/api/capabilities`). P2's wizard UI is SPA work (khelsutra repo) ? coordinate like P1b.

111 | **P2a** | **ffmpeg/ffprobe resolver** (`backend/utils/ffmpeg.py`: `ffmpeg\_bin`/`ffprobe\_bin` env-overridable via `RALLY\_FFMPEG`/`RALLY\_FFPROBE` + `require\_ffmpeg` fail-loud w/ per-OS hint) wrapping all 13 argv sites; byte-identical when unset. *(first-run cpu-mock self-test deferred to the wizard slice.)* | ? | No | ? |

[#305](https://github.com/Khelsutra/badminton-highlight-indexer/pull/305) |

112 | P2b-launcher | **Pure-Python `indexer --app` launcher** ? (Win/Mac/Linux, no native component): localhost-only bind + free-port fallback + health-poll ? `webbrowser.open` (`RALLY\_NO\_BROWSER` escape); never CLI single-shot; rides the P0e lock + P0a app-home. | P1a | No | ? |

[#303](https://github.com/Khelsutra/badminton-highlight-indexer/pull/303) |

113 | P2b-install | **Install path** (`uv tool install` / a generated cross-platform install script; ship `config.json` defaulting truly-local; optional thin per-OS shortcut to `indexer --app`) | P2b-launcher | No (signing only if clickable binaries ? 09) | ? | ? |

114 | P2c | **In-UI 3-step first-run wizard** (Gemini key / Compute / Videos), rendered from `/api/capabilities`, en/kn/hi | P1b | No | ? | ? |

115 | P2d | **DB upgrade contract** ? `SCHEMA\_VERSION` + downgrade guard (`SchemaTooNewError`) + backup-before-migrate (`<db>.bak-v<old>`, WAL-complete) + `backup\_database()` export. Byte-identical for fresh/current DBs. *(uninstall manifest deferred to P2b-install.)* | ? | No | ? |

[#307](https://github.com/Khelsutra/badminton-highlight-indexer/pull/307) |

116 | P2e | **Lazy-torch for the LIGHT tier** ? deferred the one eager `import torch` (`person\_detector.py`) so `import backend.main` works WITHOUT torch (the LIGHT-tier promise, enforced by a blocked-torch subprocess test). | ? | No | ? |

[#308](https://github.com/Khelsutra/badminton-highlight-indexer/pull/308) |

117 | **P3a** | **micromamba + conda-forge** replaces Miniconda+conda tos accept` in `deploy/cloudrun/Dockerfile` (+ ADR); reuse for the local managed WASB env | ? | No | ? | ? |

118 | P3b | **WASB weight fetcher** (SHA-256-pinned, resumable) ? `\$WASB\_WEIGHTS` ? **owner-only flag; public must NOT fetch until C3** | P3a | ? 03 (counsel) | ? | ? |

119 | P3c | **"Push to cloud" toggle** (flip cloud-serving + confirm + INR estimate + per-video cost cap) + wire worker usage-emission into the M8 ledger | P1b | Owner (cloud spend) | ? | ? |

120 | P3d | **M9 edge-auth verified path** (JWKS refresh for ~6-week rotation, Cloud Run ingress lock) | ? | ? counsel (public signup) | ? | ? |

121 | **P4a** | **ONNX export of WASB** + implement the ABC `ONNXRunner` (identical CSV) + owner byte-parity/golden gate on a real GPU ? *dissolves the dual-env* | ? | ? owner GPU | ? | ? |

|

122 | P4b | **Training-in-UI decision** (durable out-of-process long-job runner + progress/cancel, surface `gate\_verdict ship=False`) ? **recommend DEFER from v1** | ? | Owner | ? | ? |

123 | P4c | **Annotation decision** ? bundle VLC 3.0.x plugin vs build the **in-browser HTML5 annotator** emitting the same rally CSV ? **recommend in-browser**, owner-gated | ? | Owner | ? | ? |

124

125 ## 8. Open questions ? owner / external (blocking gates first)

126 1. **01 ? RESOLVED 2026-06-21:** a **webserver spun up on Windows/macOS/Linux** to power the UI is the delivery ? **no native component** (no Electron/Tauri/embedded-webview, no native-installer-as-runtime, no LGPL tray). Optional thin per-OS shortcut just runs the same command. `pystray`/LGPL stays rejected; stdlib `webbrowser` only.

127 2. **02 ? RESOLVED 2026-06-21:** v1 = **LIGHT first**, **but the sole completion goal (north star, L10) is the FULL path** ? native WASB + BYO training + in-app annotation. DoD \$9 splits "v1 shippable" from "north-star DONE"; P3/P4 are completion-gating, not optional.

128 3. **03 ? WASB-C3 counsel sign-off:** may the checkpoint be redistributed / auto-fetched into a public build at all? Determines whether a public HEAVY tier is *ever* possible vs WASB-on-cloud-only.

129 4. **04 ? Cloud in a downloaded app = hosted-Khelsutra (none built) or BYO-GCP (the only thing built)?** Decide before marketing the "push to cloud" toggle (D13 currently overstates

```

hosted).
130     5. **05 ? Training in v1 UI?** Recommend defer (synth=mock, real=GPU+dual-env,
`ship=False`).
131     6. **06 ? Annotation in v1?** in-browser HTML5 annotator (recommended) vs bundle VLC vs
defer.
132     7. **07 ? Ship AI-drafted kn/hi to real users before native-speaker review, or gate the
trilingual UI on review?**
133     8. **08 ? Shipped DEFAULT `config.json` / profile** = truly-local (so the box never
surprise-bills on Gemini nor returns 0 rallies on WASB-without-key)?
134     9. **09 ? Code-signing budget** (Windows Authenticode now, macOS Developer-ID later) so
non-tech users don't hit SmartScreen/AV prompts.
135
136     ## 9. Definition of done
137     - [ ] **01/02/08 decided in writing** (scope + delivery shape + default profile).
138     - [ ] P0a?P0e merged: launching `indexer --server` from **any** CWD finds the same
config/DB; `/api/config` leaks nothing; all CI lanes (incl. the new license-scan + built-
artifact lanes) green at the coverage floor.
139     - [ ] P1a/P1b merged: FastAPI serves the **real SPA** single-origin with refresh-
survival; capabilities/videos/reels return real data; contract test guards SPA?engine skew.
140     - [ ] P2a?P2d merged: a **clean Windows VM ? installer ? reel downloaded (en+hi)** with
**no Python/git/ffmpeg pre-installed** ; uninstall is clean; upgrade preserves data.
141     - [ ] **Every completed CUJ in the map below verified preserved** (CUJ-6/7 explicitly
owner-only and flagged, not silently dropped).
142     - [ ] Verified on a **clean Windows AND macOS AND Linux** box (the same webserver-spin-
up path on all three).
143
144     **North-star DONE (sole completion goal, L10) ? v1 is the first increment toward this,
NOT the finish line:**
145     - [ ] **Native WASB ships in the downloadable app** on Win/Mac/Linux ? via the
micromamba env and/or the **ONNX runner** (P4a) that dissolves the dual-env ? passing the owner
byte-parity gate, with **WASB-C3 cleared** (O3).
146     - [ ] **BY0 training runnable from the app** (durable out-of-process long-job runner;
`gate_verdict ship=False` surfaced honestly) ? P4b.
147     - [ ] **In-app rally annotation** (recommended: in-browser HTML5 annotator emitting the
same rally CSV) ? P4c.
148     - [ ] **CUJ-6/7 preserved in the DEFAULT download** , no longer owner-only.
149
150     ### CUJ preservation map `[verified inventory]`
151     | CUJ | Status | How preserved in the app |
152     |-----|-----|-----|
153     | **CUJ-1** Local Gemini ? rallies ? reel (en+hi) | DONE | LIGHT ships Gemini + full
`/api/process?jobs?query?compile?highlights` on the in-process worker, storage=local; verbatim,
needs key + resolved ffmpeg. |
154     | **CUJ-2** In-UI `gs://` bucket ingest | DONE | `acquire_local_input` + gcs backend
already accept `gs://` ; preserved when the user opts into a cloud storage backend (+ ADC). |
155     | **CUJ-3** Per-rally selection + query bar | DONE | Pure SPA + `/api/query` +
`/api/compile` ; preserved by bundling the built SPA single-origin + the contract test. |
156     | **CUJ-4** Reel download-to-local | DONE | `/api/compile` + `GET /api/highlights/...`
unchanged; resolved ffmpeg provides the encoder; app-home output dir keeps reels findable. |
157     | **CUJ-5** Trilingual SPA (en+kn+hi), offline | DONE | Catalogs + OFL Noto fonts
compiled into the bundle (no CDN); air-gap renderable. *(kn/hi native review pending ? 07.)* |
158     | **CUJ-6/7** Worker GCS?GPU WASB?GCS + cold-start dual-env | DONE (engine) |
**HEAVY/cloud tiers ONLY** (owner + provisioned GPU) via micromamba WASB env / Cloud Run image ?
**deliberately absent from the LIGHT default** ; gated on C3 + reproducible config. |
159     | **CUJ-8** CLI single-shot (`indexer --video-path --segmenter`) | DONE | Console +
module entrypoints untouched by packaging; available every tier (segmenter set limited by tier).
|
160     | **CUJ-9** Annotate ? train | PARTIAL | `worker train` CLI + rally-annotator CSV bridge
remain runnable; honesty fences (synth=mock, `ship=False`) must surface in any UI; recommend
**deferred** from v1 UI (P4b/c). |
161
162     ## 10. References
163     **Internal code anchors `[verified]` : ** `pyproject.toml:80-82,8-37,63-70` ;
`backend/main.py:10,64-74,144-146,275-315,646,729-736,756` ;
`backend/config/loader.py:30,112,202` ; `backend/config/presets.py` ;

```

```
`backend/compute/__init__.py`; `backend/storage/*`;
`backend/pipeline/detectors/native_wasb_runner.py:17,89,110-112`;
`backend/pipeline/stitcher/compiler.py:253`; `backend/features/rally_seq.py:17`;
`backend/infrastructure/database.py:88-110` (user_version=6); `training/gen0/harness.py:174-177`
(`ship=False`); `deploy/cloudrun/Dockerfile:33-37,54`. **Internal docs:**
`docs/COMPUTE_DECOUPLED_SERVING/`, `docs/DESKTOP_APP_PLAN.md` (superseded),
`docs/UI_PROPOSAL.md`, `docs/I18N_PLAN.md`, `kheIsutra/docs/{LOCAL_APPLIANCE_ARCHITECTURE,ANNOTATE_AND_TRAIN,DRY_RUN_CUJ,FRIENDLY_AND_MULTILINGUAL}.md`. **External `[researched]`:** uv torch-index docs; Frigate/Ollama/ComfyUI/LM Studio distribution; conda-forge/micromamba ToS; ONNX Runtime; ffmpeg GPL-vs-LGPL.
164
165     ---
166
167     ### Changelog
168     - 2026-06-21 ? Created from an exhaustive research workflow (29 agents) + per-claim re-verification at file:line.
169     - 2026-06-21 ? Owner resolved 01 + 02: **cross-platform (Win/Mac/Linux) webserver, no native component** (L9); **LIGHT-first with a WASB-complete north star as the sole completion goal** (L10). Updated $0/$2/$4.5/$6/$7-P2b/$8/$9 accordingly. Remaining gates: 03 (WASB-C3 counsel), 04 (hosted vs BYO-cloud), 05?09.
170
```

8. assistant (2026-06-22T01:56:08.613Z)

The doc is comprehensive. Now let me verify the load-bearing claims against the actual engine code on origin/master: the console-script entrypoint, the `--app` launcher, the bundled SPA serving, and lazy-torch. Let me also confirm P2b-install and P2c are still open (not merged).

9. user (2026-06-22T01:56:11.495Z)

```
=== pyproject console script ===
84:[project.scripts]
85-# `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
86-indexer = "backend.main:main"
87-
88-[tool.hatch.metadata]

=== --app flag in main.py ===
origin/master:backend/main.py:131:# --- P2: the friendly `indexer --app` launcher (cross-
platform, no native component ? 01) ---
origin/master:backend/main.py:152:     """For --app: keep `port` if free, else fall back to a
free one so a busy 8000 never blocks the app."""
origin/master:backend/main.py:867:     parser.add_argument("--app", action="store_true",
origin/master:backend/main.py:981:     # CLI processing mode (no web UI needed). --app/--server
always run the server, never CLI single-shot.
origin/master:backend/main.py:1074:     # with --app, or the poll 400s and the browser
just won't auto-open ? the server still runs).

=== run_app / launcher function ===
```

10. user (2026-06-22T01:56:13.682Z)

```
=== bundled SPA serving (backend/ui) ===
origin/master:backend/main.py:184:# (The legacy in-engine dashboard; the real SPA gets bundled
here in P1a.)
origin/master:backend/main.py:194:# P1b: serve the bundled KheIsutra SPA single-origin.
`backend/ui/` is the committed snapshot
origin/master:backend/main.py:196:# absent (a source checkout with no vendored UI). The SPA
calls same-origin `/api`, so no SPA change
origin/master:backend/main.py:198:# (no client-side path routing ? no history-fallback catch-
all, which would also be a route-order hazard).
origin/master:backend/main.py:215:def _bundled_ui_version() -> Optional[Dict[str, Any]]:
origin/master:backend/main.py:216:     """Provenance of the vendored SPA
(backend/ui/ui_version.json), or None if no UI is bundled.
```

```

origin/master:backend/main.py:218: path = os.path.join(_BUNDLED_UI_DIR, "ui_version.json")
origin/master:backend/main.py:263: # The UI locale the SPA was rendered in (e.g. "kn", "hi-
IN"). Recorded on the job for PMF
origin/master:backend/main.py:278: # The UI locale the SPA was rendered in (e.g. "kn", "hi-
IN"). Localizes the reel watermark
origin/master:backend/main.py:308: """Cheap liveness/readiness probe (no auth) for the SPA's
offline banner
origin/master:backend/main.py:320:# P1: the bundled SPA pins this ? the contract version of the
/api surface. BUMP when an existing
origin/master:backend/main.py:335: """Version handshake so a bundled SPA can detect engine
skew (PACKAGING_PLAN.md P1)."""
origin/master:backend/main.py:336: return {"engine_version": _engine_version(),
"api_version": API_VERSION, "ui": _bundled_ui_version()}
origin/master:backend/main.py:751: # the UI locale the SPA rendered in (req.locale): the
reel's watermark reads in the user's language
origin/master:backend/main.py:766: # download URL so the SPA needn't hand-build it (P6 /
PER_RALLY_SELECTION.md §4.3).

```

=== backend/ui dir on master ===

```

100644 blob 6c4ef042e0f7f1b0aec0c4eb3e8129856f467189 backend/ui/THIRD_PARTY_NOTICES.md
040000 tree bfb230dbe062f537128f5a30d6e53cbb62b70078 backend/ui/assets
100644 blob 61378a9640e76b8f598ed1dfc45ef874e4f01c51 backend/ui/index.html
100644 blob 51a9859f6f0e8e653c7bfcd0ca76ab070f17fc97 backend/ui/ui_version.json

```

=== lazy torch test ===

```

origin/master:backend/pipeline/detectors/person_detector.py:1: """Person-detection rally cue ?
torchvision detector + ByteTrack + motion windowing.
origin/master:backend/pipeline/detectors/person_detector.py:11: - `lazy_load_model`
? load a permissively-licensed torchvision detector
origin/master:backend/pipeline/detectors/person_detector.py:18:The model libs (torch,
torchvision, supervision) are imported lazily inside the methods, and the
origin/master:backend/pipeline/detectors/person_detector.py:20:registry never loads a detector
model NOR requires torch (the LIGHT tier ships no torch ? P2e;
origin/master:backend/pipeline/detectors/person_detector.py:23:Licensing: torchvision detectors
are BSD + COCO weights; ByteTrack is MIT

```

11. assistant (2026-06-22T01:56:17.984Z)

All core claims verify. Now let me confirm the `--app` launcher actually boots the server + opens the browser (the function body), and confirm the bundled UI is actually mounted at `/.`. Let me also double-check P2b-install and P2c are genuinely not yet shipped.

12. user (2026-06-22T01:56:19.132Z)

```

=== --app launcher body (main.py ~1040-1080) ===
    return

```

```

    print(f"[CLI] Indexing play segments using segmenter: {seg_name}...")
    segmenter_actual = seg_name
    play_wm, cand_wm = db_instance.segment_watermarks(video_id)
    segmenter = segmenter_cls(db_instance, global_config)
    result = segmenter.process_video(video_id, local_video, max_frames=args.max_frames)
    segmentation_ran = True

    if isinstance(result, Failure):
        print(f"[FAIL] Segmenter '{seg_name}' failed: {result.message}")
        print(f"[FAIL] Indexing halted for video.")
        segments = []
    else:
        segments = result.value if hasattr(result, "value") else result
        print(f"[SUCCESS] Process complete. Indexed {len(segments)} play segments inside
database: {db_path}")
        # Destroy-AFTER-confirm: keep new on success, restore prior on empty/failure.
        _finalize_reingest(video_id, segments, play_wm, cand_wm)
    finally:

```

```

        # Always emit the per-video observability report. Never raises.
        paths = emit_indexer_report(
            db=db_instance, video_id=video_id, video_path=args.video_path,
config=global_config,
            val_results=val_results, val_context=val_context,
            segmenter_requested=args.segmenter, segmenter_actual=segmenter_actual,
            was_reingest=was_reingest, segmentation_ran=segmentation_ran,
        )
        if paths:
            print(f"[REPORT] Technical report : {paths.get('technical_md')}")
            print(f"[REPORT] Customer summary : {paths.get('customer_md')}")
            print(f"[REPORT] Machine JSON      : {paths.get('json')}")
        return

# Start Local Server Mode
if args.app or args.server or not args.video_path:
    if args.app:
        # P2: the friendly launcher ? always localhost-only (use --server for network
exposure),
        # auto-pick a free port if --port is busy, and open the bundled UI once the server
is up.
        bind_host = "127.0.0.1"
        port = _resolve_app_port(bind_host, args.port)
        if port != args.port:
            print(f"[APP] Port {args.port} is busy ? using {port} instead.")
        url = f"http://{bind_host}:{port}/"
        # The browser-open health poll sends Host: 127.0.0.1 ? it relies on loopback staying
in
        # the Host allowlist (the default; don't set RALLY_ALLOWED_HOSTS to a non-loopback-
only set
        # with --app, or the poll 400s and the browser just won't auto-open ? the server
still runs).
        print(f"[APP] Starting KhelSutra ? opening {url} in your browser "
            f"(set RALLY_NO_BROWSER=1 to skip). Press Ctrl+C to stop.")
        threading.Thread(target=_open_browser_when_ready, args=(url, url + "api/health"),
            daemon=True).start()
    else:
        bind_host = _resolve_bind_host(global_config, args.host) # P0c: config/profile-
driven (default 127.0.0.1)
        port = args.port
        print(f"[SERVER] Starting local FastAPI server on http://{bind_host}:{port}...")
        if bind_host not in ("127.0.0.1", "localhost", "::1"):
            print(f"[SERVER] ? Binding {bind_host} exposes the API on the network ? ensure
an auth "
                f"gate (edge_auth_enabled / a reverse proxy) is in front, and set ")

```

13. user (2026-06-22T01:56:21.916Z)

=== StaticFiles mount of backend/ui ===

P1b: serve the bundled KhelSutra SPA single-origin. `backend/ui/` is the committed source (produced by scripts/bundle_ui.sh); fall back to the legacy in-engine dashboard only if absent (a source checkout with no vendored UI). The SPA calls same-origin `/api`, so no auth is needed; it is query-param driven (?video=?), so serving index.html at `/` + /assets is needed (no client-side path routing ? no history-fallback catch-all, which would also be a route).

```

_BUNDLED_UI_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "ui")
_UI_DIR = _BUNDLED_UI_DIR if os.path.isfile(os.path.join(_BUNDLED_UI_DIR, "index.html")) else
_STATIC_DIR
_UI_ASSETS_DIR = os.path.join(_UI_DIR, "assets")
if os.path.isdir(_UI_ASSETS_DIR):
    app.mount("/assets", StaticFiles(directory=_UI_ASSETS_DIR), name="assets")

```

```

@app.get("/", response_class=HTMLResponse)
def serve_index():
    index_path = os.path.join(_UI_DIR, "index.html")
    if os.path.exists(index_path):
        with open(index_path, "r", encoding="utf-8") as f:
            return f.read()
    return "<h3>Sports Highlight Indexer Server is Running. (No bundled UI found.)</h3>"

def _bundled_ui_version() -> Optional[Dict[str, Any]]:
    """Provenance of the vendored SPA (backend/ui/ui_version.json), or None if no UI is bundled.
    Includes `engine_api_version` ? the API_VERSION the snapshot was built against (skew
    guard)."""
    path = os.path.join(_BUNDLED_UI_DIR, "ui_version.json")
    try:
        with open(path, "r", encoding="utf-8") as f:
            return json.load(f)
    except (OSError, ValueError):
        return None

```

Global variables initialized at startup

```

db_instance: Optional[Database] = None
global_config: Optional[AppConfig] = None # now typed (Pydantic model)

```

P0e: the single-instance lock handle. Kept module-global for the process lifetime ? if i
GC'd/closed the OS would release the lock. None until main() acquires it (server/CLI in

```

_instance_lock: Optional[Any] = None

```

Async job worker ? drain/wake loop (DEFERRED_HARDENING_PLAN #16, EXECUTIO
now lives in backend.api.job_worker. Re-exported here so the names stay addressabl
`backend.main` (`JobWaiting`, `\_drain\_due\_jobs`, `\_arm\_wake\_timer`, `\_get\_executor`
`\_run\_claimed\_job`, `\_MAX\_DRAIN\_WAKE\_SEC`, ...). The worker resolves these seam
this module object at call-time, so `monkeypatch.setattr(backend.main, ...)` still
intercepts the inter-function calls (a parked job arms the PATCHED `_arm_wake_timer
so no real daemon timer leaks in tests). Behavior is unchanged by the extraction.

```

from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable on
    backend.main)
    JobWaiting,

```

```

=== search for P2b-install / install script / uv tool install in repo ===
origin/master:docs/PACKAGING_PLAN.md

```

14. assistant (2026-06-22T01:56:27.172Z)

Confirmed: the `--app` launcher (P2b-launcher) is fully wired ? boots localhost server, free-
port fallback, health-poll, `webbrowser.open`. The bundled SPA serves at `/' + `'/assets`. And
the only place `uv tool install`/install-script appears is the PLANNING doc itself (no actual
install script artifact shipped), confirming P2b-install is still open.

Let me verify P2c (first-run wizard) hasn't shipped in the khelsutra SPA, and check the latest
khelsutra log for anything wizard-related.

15. user (2026-06-22T01:56:30.379Z)

```

---KHELSUTRA LOG---

```

```

17242a9 feat(site): localize own-model body content to kn/hi (#86)

```

```

7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)

```

5e5a2ac docs(friendly-multilingual): close out the chapter (§9 DoD + UI/UX + B7 deferred) (#84)
 4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero (#83)
 0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
 1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
 64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit (#80)
 8328721 docs(tracker): refresh the §7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
 3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook (RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
 9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
 9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
 aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
 1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)
 c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)
 1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)
 f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru (#71)
 bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling tracked (#301) (#69)
 46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
 a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
 ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
 4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.* tunnel + CF Access) (#65)
 8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin /api) (#64)
 92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
 501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*) (#62)
 95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)

16. user (2026-06-22T01:56:30.547Z)

No files found

17. assistant (2026-06-22T01:56:36.230Z)

Confirmed: no wizard/first-run/capabilities/onboarding code in the khelsutra SPA ? P2c is genuinely open. The khelsutra log shows no wizard work; the SPA has upload (T4 #66) but no first-run wizard rendered from `/api/capabilities`.

Let me do one final cross-check: confirm the built-wheel CI lane exists (so "wheel" is a real verified artifact path, not just a doc claim) and check whether P3 GPU validation actually landed (the memory says "P3 DONE 2026-06-22" but the doc lists it as #1 remaining ? I need to reconcile this).

18. user (2026-06-22T01:56:40.691Z)

=== built-wheel CI lane (P0d) ===
 origin/master:.github/workflows/ci.yml

=== any P3 validation commit / screencast / 22 rallies in engine log ===
 fefa412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)

=== P3 status row in doc (re-grep) ===
 15:1. **P3 GPU validation + screencast (owner wants first; ~\$0.50, well under the \$100 cap).**
 Provision an L4 (`deploy/gcp/create\_gpu\_vm.sh`) ? install the **wheel** on the fresh GPU box + the WASB env ? validate the packaged stack end-to-end (`indexer --app` ? bundled SPA ? native-WASB ? reel) ? **screencast** ? `teardown.sh` (deletes) + confirm \$0. ? Run as a FOCUSED op (billable VM ? teardown is priority #1). ? Known blocker: native-WASB returned **0 rallies** on

a fresh build last session ? reproducing the 22-rally result needs the engine session's exact tuned `worker infer detector\_impl=native` config. Confirm cmd + cost before spend; `[[gcp-vm-always-delete]]`.

19. assistant (2026-06-22T01:56:45.694Z)

Important reconciliation: the MEMORY index claims "? P3 GPU validation DONE (2026-06-22) ? 22 rallies ? 73.7 MB reel" but the **committed `PACKAGING\_PLAN.md` on origin/master still lists P3 as remaining item #1** with a known blocker (native-WASB returned 0 rallies on a fresh build, needs the tuned config to reproduce 22 rallies). There's no engine commit for a P3 GPU-validation screencast. So the doc (git-grounded) and the memory disagree ? I'll trust git/the doc and flag this.

Let me confirm the built-wheel smoke lane content in CI to nail down the "wheel works" claim, then I have everything.

20. user (2026-06-22T01:56:46.997Z)

fatal: ambiguous argument 'origin\master;.github\workflows\ci.yml': unknown revision or path not in the working tree.

Use '--' to separate paths from revisions, like this:

'git <command> [<revision>...] -- [<file>...]'

21. user (2026-06-22T01:56:59.916Z)

fatal: ambiguous argument 'origin\master;.github\workflows\ci.yml': unknown revision or path not in the working tree.

Use '--' to separate paths from revisions, like this:

'git <command> [<revision>...] -- [<file>...]'

22. assistant (2026-06-22T01:57:02.548Z)

The PowerShell-vs-Bash path munging is interfering. Let me use the Bash tool with a heredoc-free single quote approach.

23. user (2026-06-22T01:57:04.487Z)

(Bash completed with no output)

24. user (2026-06-22T01:57:08.579Z)

fatal: ambiguous argument 'origin\master;.github\workflows\ci.yml': unknown revision or path not in the working tree.

Use '--' to separate paths from revisions, like this:

'git <command> [<revision>...] -- [<file>...]'

===SIZE===

3 /tmp/ci.yml

===workflow files===

100644 blob 89cea611236623f978c01ed2ff015ca01224bb39 .github/workflows/ci.yml

25. assistant (2026-06-22T01:57:17.944Z)

The earlier `git grep` already confirmed `ci.yml` contains the `indexer --help`/built-wheel/pip-licenses patterns (P0d). That's sufficient grounding ? the wheel-build + license-scan lanes exist on master. I have enough to answer precisely. Let me record this verification finding for the parent.

The git evidence is complete. Summary of what's verified:

****Works TODAY (verified on origin/master):****

- Console script `indexer = "backend.main:main"` (pyproject.toml:86)
- `indexer --app` launcher (main.py:867, body ~1065-1078): localhost-only bind, free-port fallback, health-poll, `webbrowser.open`, `RALLY\_NO\_BROWSER` escape
- Bundled SPA at `backend/ui/` (index.html + assets + ui_version.json + THIRD_PARTY_NOTICES)

committed), served at `` and `/assets` (main.py:194-210)
- Lazy-torch LIGHT tier (person_detector.py:18-23, P2e #308)
- Built-wheel smoke + license-scan CI lanes (ci.yml, P0d #297)

****Open gaps (verified NOT merged):**** P2b-install (no install script artifact anywhere except the doc), P2c first-run wizard (no wizard/capabilities code in khelsutra SPA).

****Stale-claim flag:**** MEMORY index says P3 GPU validation is DONE; the git-committed doc + engine log show it is NOT ? still remaining item #1 with a known 0-rallies blocker.

26. user (2026-06-22T01:57:55.728Z)

Structured output provided successfully

27. assistant (2026-06-22T01:58:05.416Z)

Verification complete. I scanned `docs/PACKAGING\_PLAN.md` end-to-end on engine `origin/master` and cross-checked every load-bearing claim against the actual code and git log. Findings returned via StructuredOutput.

Key answers to the co-founder's question "can I download a package and run it on a new machine now?":

1. ****What works today**** ? Yes for a developer: `pip install` the wheel, then `indexer --app`. Verified on origin/master: console entrypoint `indexer = backend.main:main` (pyproject.toml:86), the `--app` launcher (main.py:867 + body) boots a localhost server, picks a free port, health-polls, and opens the browser to the bundled SPA, which is committed at `backend/ui/` and served at `` (P1a #300, P1b #302). The LIGHT tier is torch-free (P2e #308). There is ****no double-click installer**** ? Python + a terminal command are still required.

2. ****Exact remaining gap to a non-technical user**** ? both git-verified as NOT merged: ****P2b-install**** (no install script/`uv tool install`/default config/uninstall ? strings appear only in the plan doc) and ****P2c first-run wizard**** (zero wizard/capabilities/onboard matches in khelsutra/web; the SPA has upload but no guided Gemini-key/compute step).

3. ****One stale-doc flag worth surfacing****: the MEMORY index says "P3 GPU validation DONE (22 rallies ? 73.7 MB reel)", but the committed `PACKAGING\_PLAN.md` still lists P3 as remaining item #1 with a known 0-rallies blocker, and there's no validation commit on master. The memory and git disagree ? trust git and reconcile with the owner before spending on the L4.